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1. INTRODUCTION

1.1 Project Overview

The Gemini Historical Artifact Analyzer is a web-based application designed to analyze images of historical artifacts using advanced Generative AI technology. The system integrates Google's Gemini API, which supports multimodal input (text and image), to generate context-aware and informative descriptions.

The application workflow includes image upload, image preprocessing, API interaction, and AI-generated response display. When a user uploads an artifact image, the system encodes the image and sends it to the Gemini model along with a structured prompt. The AI model processes both the image and text instructions to generate a detailed explanation covering:

- Historical period
- Cultural significance
- Possible origin
- Interesting and educational facts

The project is implemented using Python, Streamlit for the user interface, and the Google GenAI SDK for AI integration. The system demonstrates the practical application of Artificial Intelligence in historical research and educational support.

1.2 Purpose

The primary purpose of this project is to develop an intelligent system that can automatically analyze and describe historical artifacts using Artificial Intelligence. The project aims to simplify historical learning by providing instant and structured insights about artifacts based on visual input.

This system is designed to:

- Provide automated artifact analysis using AI
- Enhance historical learning through interactive technology
- Reduce dependency on manual research for basic artifact understanding
- Demonstrate the integration of multimodal AI models in real-world applications

By combining image processing and generative AI, the project showcases how modern AI systems can assist in educational and cultural domains, making historical knowledge more accessible and engaging.

2. IDEATION PHASE

2.1 Problem Statement

Date	22 February 2026
Team ID	LTVIP2026TMIDS70177
Project Name	Gemini Historical Artifact Description
Maximum Marks	2 Marks

Understanding historical artifacts requires domain expertise, contextual knowledge, and access to reliable historical sources. Students, researchers, and history enthusiasts often face difficulties in identifying the historical period, cultural significance, and origin of artifacts based solely on images. Traditional research methods involve manual searching, reading multiple sources, and consulting experts, which can be time-consuming and inefficient.

Therefore, there is a need for an intelligent system that can automatically analyze images of historical artifacts and generate meaningful, structured, and educational descriptions using Artificial Intelligence. The proposed system aims to solve this problem by leveraging Google's Gemini multimodal AI model to provide instant artifact analysis based on image input.

2.2 Empathy Map Canvas

Date	22 February 2026
Team ID	LTVIP2026TMIDS70177
Project Name	Gemini Historical Artifact Description
Maximum Marks	4 Marks

Target User:

Students, history learners, researchers, museum visitors, and educators who want to understand historical artifacts using images.

THINK & FEEL

- Curious about the history behind artifacts.
 - Wants accurate and trustworthy information.
 - Feels excited when discovering new historical knowledge.
 - Sometimes confused about identifying artifact origin or period.
 - Wants quick and structured explanations without complex research.
-

SEE

- Historical artifacts in museums, books, and online platforms.
- Limited information boards near artifacts.
- Complex research articles with difficult terminology.
- General image search results that lack structured details.

2.3 Brainstorming

Date	22 February 2026	
Team ID	LTVIP2026TMIDS70177	
Project Name	Gemini Historical Artifact Description	
Maximum Marks	4 Marks	

- ☐ How might we automatically analyze historical artifact images using AI?
- ☐ How might we generate structured descriptions including historical period and cultural significance?
- ☐ How might we integrate Google Gemini’s multimodal capabilities into a web-based application?
- ☐ How might we improve user learning experience using AI-generated insights?
- ☐ How might we ensure accurate and reliable historical interpretation?
- ☐ How might we design a simple and user-friendly interface for image uploads?

Brainstorming Insights

During the brainstorming phase, the team explored multiple approaches to build an AI-powered artifact analysis system. The discussion focused on integrating image processing with large language models to generate structured historical explanations. The team identified the need for:

- A multimodal AI model capable of understanding both text and images.
- A simple web interface for uploading artifact images.
- A structured output format including historical period, origin, cultural importance, and interesting facts.
- A scalable and cloud-based solution for real-time processing.
- Future expansion possibilities such as multilingual support and database integration.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The customer journey map for the *Gemini Historical Artifact Description System* illustrates the end-to-end experience of a user interacting with the AI-powered artifact analysis platform. The journey begins when a user encounters a historical artifact and seeks to understand its background. Instead of manually researching through books or websites, the user uploads an image into the system.

The system processes the image using Google Gemini’s multimodal AI model and generates a structured historical description including the artifact’s period, origin, cultural significance, and interesting facts. The journey concludes when the user gains meaningful historical insights in real time, enhancing their learning experience.

This journey map aims to identify user needs, pain points, and opportunities at each stage to ensure an intuitive, efficient, and educational experience.

3.2 Solution Requirement

Date	22 February 2026
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Maximum Marks	4 Marks

Functional Requirements:

The solution for the *Gemini Historical Artifact Description System* must deliver an AI-powered, web-based application that analyzes historical artifact images and generates structured descriptions. The system should provide real-time insights using Google Gemini’s multimodal AI capabilities.

The system must:

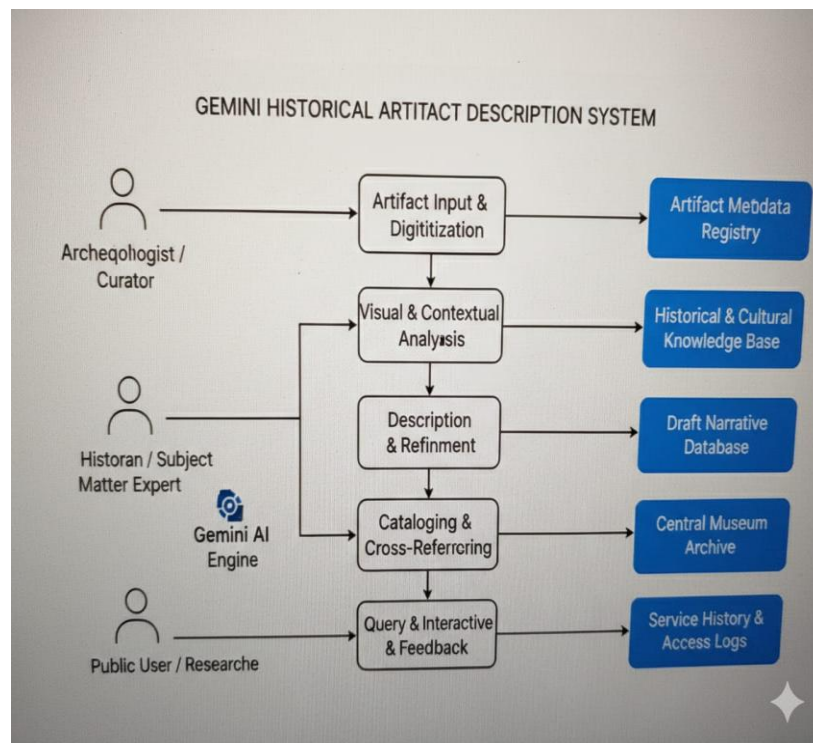
1. Allow users to upload images of historical artifacts through a web interface.
2. Accept optional text input to guide the analysis.
3. Process both image and text inputs using Google Gemini API.
4. Generate structured historical descriptions including:
 - Historical period
 - Cultural significance
 - Possible origin
 - Interesting facts
5. Display AI-generated results clearly and in a user-friendly format.
6. Provide real-time response after submission.
7. Handle errors gracefully (e.g., no image uploaded).
8. Ensure secure API key usage through environment variables.
9. Be scalable and accessible through web browsers.
10. Support future enhancements such as multilingual output and artifact database integration.

3.3 Data Flow Diagram

Date	22 February 2026
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Maximum Marks	4 Marks

The Data Flow Diagram (DFD) for the *Gemini Historical Artifact Description System* illustrates how data moves between the user, the web application, and the Gemini AI model. It demonstrates how image inputs are processed and transformed into structured historical descriptions.

The system captures user-uploaded images, sends them to the Gemini API for analysis, processes the AI response, and displays the results back to the user. The DFD ensures clarity in understanding how input data flows through the system and produces meaningful output.



3.4 Technology Stack

Date	22 February 2026
Team ID	LTVIP2026TMIDS70177
Project Name	Gemini Historical Artifact Description
Maximum Marks	4 Marks

The technology stack for the *Gemini Historical Artifact Description System* combines modern web technologies with advanced AI capabilities to deliver a scalable, intelligent, and user-friendly solution. The system leverages Python-based backend development, Streamlit for web interface creation, and Google Gemini’s multimodal AI model for image and text processing.

This stack ensures real-time artifact analysis, secure API integration, and seamless user interaction.

Technology Stack Table

Layer	Technology	Purpose
Frontend	Streamlit	Builds interactive web interface for image upload and result display
Backend	Python	Handles application logic and API integration
AI Model	Google Gemini (Multimodal API)	Analyzes image + text input and generates structured historical descriptions
Environment Management	Python Virtual Environment (venv)	Isolates project dependencies
API Security	.env + python-dotenv	Securely stores and loads API keys
Image Processing	PIL (Python Imaging Library)	Handles uploaded image data
Package Management	pip	Installs and manages required libraries

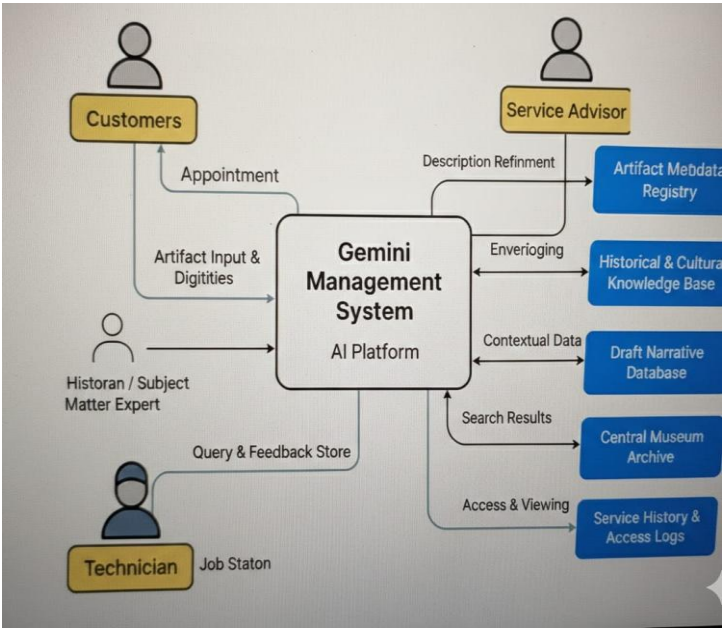
4. PROJECT DESIGN

4.1 Problem Solution Fit

Date	22 February 2026
Team ID	LTVIP2026TMIDS70177
Project Name	Gemini Historical Artifact Description
Maximum Marks	4 Marks

The primary problem identified is the difficulty in analyzing and understanding historical artifacts based solely on visual observation. Students, researchers, and history enthusiasts often lack immediate access to expert interpretation, leading to time-consuming manual research and incomplete understanding.

The proposed solution, the *Gemini Historical Artifact Description System*, effectively addresses this challenge by integrating a multimodal AI model capable of analyzing both image and text inputs. By leveraging Google Gemini’s advanced capabilities, the system generates structured historical descriptions including the artifact’s historical period, cultural significance, possible origin, and interesting facts.



4.2 Proposed Solution

The proposed solution, *Gemini Historical Artifact Description System*, is an AI-powered web application designed to analyze historical artifact images and generate structured descriptions in real time. The system leverages Google Gemini's multimodal AI capabilities to interpret both visual and textual inputs.

The solution provides a simple and interactive interface where users can upload an image of a historical artifact. The backend processes the image and sends it to the Gemini API along with a structured prompt. The AI model then generates detailed historical insights including the artifact's historical period, cultural significance, possible origin, and interesting facts.

The application is designed to be user-friendly, scalable, and secure, ensuring efficient API handling and real-time response generation.

Key Components of the Proposed Solution

1. Image Upload & User Interface

- Web-based interface built using Streamlit.
 - Allows users to upload artifact images.
-

2. AI-Based Image & Text Processing

- Uses Google Gemini multimodal model.
 - Processes both image and textual prompts.
-

3. Backend Processing Logic

- Python-based backend handles API integration.
 - Converts uploaded images into appropriate format.
-

4. Structured Output Display

- Displays analysis results clearly.
 - Sections include:
 - Historical Period
 - Cultural Significance
 - Possible Origin
-

5. Scalability & Future Enhancements

- Can be extended to support:
 - Multilingual output
 - Artifact database integration

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

5.1 Project Planning

The project planning for the *Gemini Historical Artifact Description System* focuses on developing a structured, AI-powered web application that provides real-time historical artifact analysis. The project follows a systematic and phased development approach to ensure proper requirement gathering, efficient implementation, testing, and successful deployment.

The planning strategy emphasizes clear milestone definition, risk management, iterative development, and quality assurance to deliver a reliable and scalable AI-based solution.

1. Project Phases & Timeline

Phase 1: Requirement Analysis & Research

- Understanding the problem statement
- Identifying user needs (students, researchers, historians)

Phase 2: System Design

- Designing application architecture
- Creating data flow diagrams

Phase 3: Development & Implementation

- Setting up Python virtual environment
- Installing required libraries (streamlit, google-generativeai, dotenv, PIL)
- Implementing image upload functionality

Phase 4: Testing & Debugging

- Unit testing individual modules
- API response validation
- Handling runtime errors and exceptions

Phase 5: Deployment & Documentation

- Running application via Streamlit
- Final UI validation
- Preparing project documentation

Phase 6: Future Enhancement Planning

- Adding multilingual support
- Exporting results as PDF

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Date	22 February 2026
Team ID	LTVIP2026TMIDS70177
Project Name	Gemini Historical Artifact Description
Maximum Marks	4 Marks

Performance Testing Overview

Functional and performance testing for the *Gemini Historical Artifact Description System* ensures that all components of the application operate correctly, efficiently, and reliably under various usage conditions. The testing process validates image upload functionality, API communication, response accuracy, system stability, and real-time performance.

The primary objective is to deliver a responsive, stable, and scalable AI-based web application capable of handling multiple artifact image analyses without failure.

1. Functional Testing

Functional testing focuses on verifying that each module performs as expected.

- Image upload validation (supported formats: JPG, PNG)
- API key authentication and secure loading

2. Performance Testing

Performance testing evaluates system behavior under different load conditions.

- Response time measurement for API calls
- Testing with multiple image uploads

3. API Response Testing

- Validating structured output format
- Checking completeness of generated descriptions

4. Security Testing

- Secure storage of API key using .env
- Preventing exposure of sensitive credentials

Tools & Techniques Used

- Manual Testing for UI validation
- Streamlit Runtime Logs for debugging
- Python Exception Handling
- API Response Monitoring
- Virtual Environment (venv) for dependency isolation

Testing Outcome

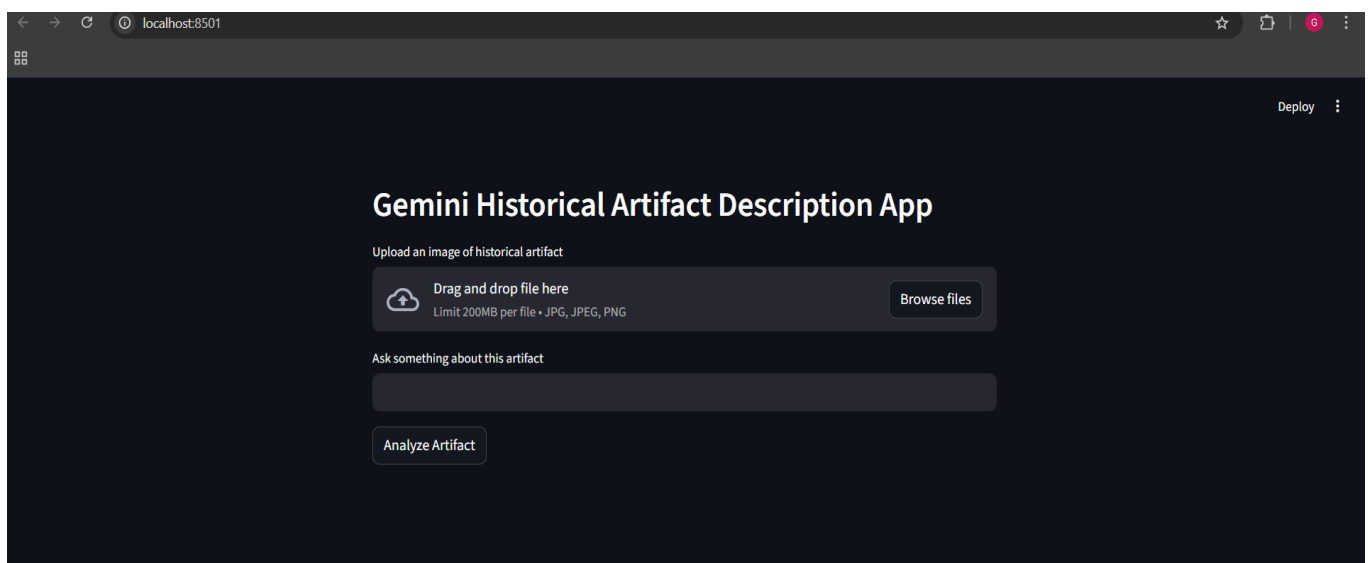
The system successfully:

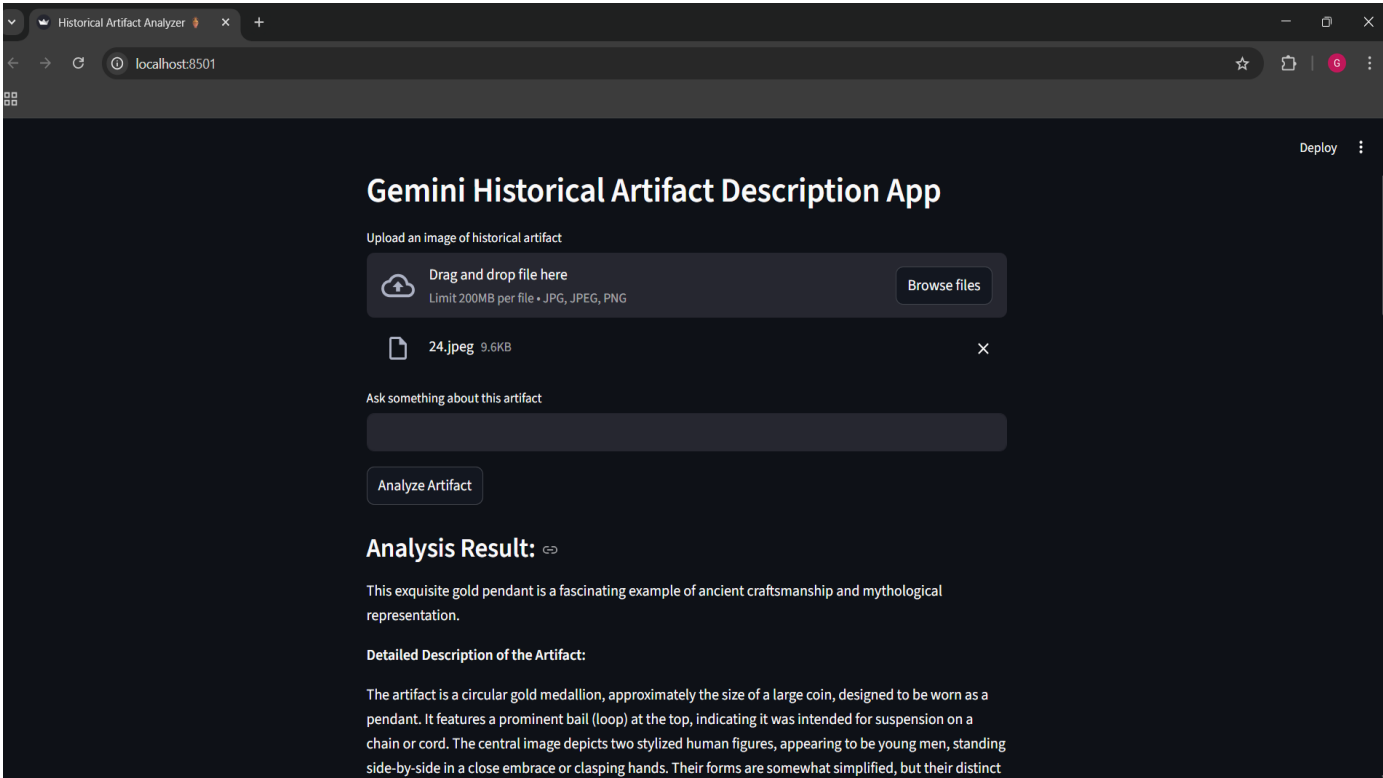
- Generates real-time historical descriptions
- Maintains stable performance under normal usage
- Handles errors gracefully

This confirms that the application meets functional and performance expectations.

7. RESULTS

7.1 Output Screenshots:





Analysis Result:

This exquisite gold pendant is a fascinating example of ancient craftsmanship and mythological representation.

Detailed Description of the Artifact:

The artifact is a circular gold medallion, approximately the size of a large coin, designed to be worn as a pendant. It features a prominent bail (loop) at the top, indicating it was intended for suspension on a chain or cord. The central image depicts two stylized human figures, appearing to be young men, standing side-by-side in a close embrace or clasping hands. Their forms are somewhat simplified, but their distinct heads, torsos, and legs are clearly visible. They face forward, their expressions serene.

The central scene is enclosed within a raised border. Surrounding this border, there is a band of small, regularly spaced dots or beads, followed by an outer rim decorated with radiating lines or grooves, reminiscent of sun rays or a stylized laurel wreath. The overall impression is one of elegance and symbolic depth, crafted from precious gold, signifying wealth and importance.

Historical Period:

Based on the style of the figures, the iconography, and the use of gold for such a medallion, this artifact most likely dates to **Classical Antiquity, specifically the Hellenistic period (c. 323–31 BCE) or the Roman Imperial period (c. 27 BCE – 476 CE)**. The artistic style is characteristic of Greco-Roman art, which frequently depicted mythological figures in jewelry and personal adornments. Gold jewelry was highly prized and widely produced across the vast Roman Empire and its Hellenistic predecessors.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- 1. AI-Powered Analysis**
Automatically analyzes artifact images using advanced multimodal AI, reducing manual research effort.
- 2. Real-Time Results**
Generates structured historical descriptions instantly after image upload.
- 3. User-Friendly Interface**
Built using Streamlit, making it simple and easy for students, researchers, and historians to use.

4. Multimodal Capability

Processes both image and text inputs for more accurate and detailed descriptions.

5. Time Efficient

Saves significant time compared to traditional manual historical research methods.

6. Scalable Architecture

Can be extended with multilingual support, PDF export, or museum database integration.

7. Secure API Handling

Uses environment variables (.env) to protect API keys and sensitive data.

8. Educational Support

Useful for students, museums, researchers, and cultural heritage institutions.

Disadvantages:

1. Dependency on Internet

Requires stable internet connection to access the Gemini API.

2. API Cost Limitations

Continuous usage may incur API usage costs.

3. AI Interpretation Limitations

The model may sometimes provide generalized or partially inaccurate interpretations.

4. No Physical Verification

Cannot physically examine artifacts like a human expert.

5. Limited Context Understanding

If the image quality is poor, results may not be accurate.

6. Data Privacy Concerns

Uploaded images are processed through an external API.

7. Performance Delay

Large images or slow networks may increase response time.

7.CONCLUSION

The *Gemini Historical Artifact Description System* successfully demonstrates how artificial intelligence can be applied to simplify and enhance historical artifact analysis. By integrating Google Gemini's multimodal AI capabilities with a user-friendly Streamlit interface, the system enables users to upload artifact images and receive structured historical descriptions in real time.

The project effectively addresses the challenge of limited access to expert historical interpretation by providing automated insights such as historical period, cultural significance, possible origin, and interesting facts. The implementation ensures secure API integration, efficient backend processing, and reliable performance.

Through proper planning, design, development, and testing, the system achieves its objective of delivering an intelligent, scalable, and easy-to-use solution. This project highlights the potential of AI-powered applications in education, research, and cultural heritage preservation.

Overall, the system proves that combining multimodal AI with web technologies can create innovative tools that make knowledge more accessible, interactive, and efficient.

10. FUTURE SCOPE

The *Gemini Historical Artifact Description System* has significant potential for further enhancement and expansion. While the current version provides real-time AI-based artifact analysis, several improvements can make the system more advanced, scalable, and impactful.

1. Multilingual Support

The system can be enhanced to generate descriptions in multiple languages, making it accessible to global users, researchers, and museums.

2. Artifact Database Integration

A structured database can be integrated to store previously analyzed artifacts. This would allow users to search, compare, and retrieve historical records efficiently.

3. PDF & Report Generation

The system can include an option to download generated descriptions as PDF reports for academic submissions or museum documentation.

4. Advanced Image Recognition

Future improvements may include better object detection and classification to identify artifact categories automatically (e.g., coins, sculptures, manuscripts).

5. Mobile Application Development

The project can be extended into a mobile application for easier accessibility and field usage in museums or archaeological sites.

6. AR/VR Integration

Augmented Reality (AR) or Virtual Reality (VR) integration could allow users to visualize artifacts in 3D with interactive historical insights.

7. Integration with Museum Systems

The system can be connected with museum management systems for automated cataloging and digital archiving .

8. Improved Accuracy with Fine-Tuning

Future versions can improve accuracy by refining prompts or using domain-specific datasets for better historical interpretation.

These future enhancements can transform the system into a comprehensive AI-powered digital heritage assistant, supporting education, research, and cultural preservation on a larger scale.

11. APPENDIX

GitHub :- <https://github.com/ganeshch25/Gemini-Historical-Articraft-Description>

Project Demonstration Link :-

https://drive.google.com/file/d/1r4UfoPh7V_yyrUomsiTolTuYrPDlsJgQ/view?usp=drive_link